Quantum Gravity from Recognition Science: EFE Emergence, Equivalence Principle, and Gravitational Waves

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Abstract

We present the RS theory of gravity. The Einstein Field Equations (EFEs) $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ emerge from RS action stationarity with coupling constant $\kappa = 8\pi G/c^4 = \pi\lambda_{\text{rec}}^2/(\hbar c)$ derived from the recognition wavelength. The equivalence principle holds exactly: all masses arise from J-cost, so inertial = gravitational mass by construction. The graviton does not exist as a separate quantum — gravity is emergent curvature. Galaxy rotation curves are explained by the ILG mechanism without dark matter particles. Gravitational waves propagate at exactly c because they are disturbances in the same ledger that defines c. All zero free parameters.

1 Introduction

General Relativity (GR) describes gravity as spacetime curvature, governed by the Einstein Field Equations. Quantum gravity—reconciling GR with quantum mechanics—is one of the deepest open problems in physics.

RS [1] dissolves rather than solves the quantum gravity problem: gravity is not a fundamental force requiring separate quantization. It is emergent curvature of the discrete recognition ledger. When matter-energy (ledger entries) curve the ledger metric, what looks like a gravitational force is a geodesic deviation. The "quantization" is already implicit: the ledger is fundamentally discrete, and the EFEs emerge in the continuum limit. The RS approach therefore does not produce a graviton as a fundamental particle; instead, gravitational wave quanta (if detectable) are collective excitations of the ledger metric.

The RS axioms [1] are:

(A1) $J(1) = 0$ *(Normalization)*

(A2) $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$ *(Recognition Composition Law)*

(A3) $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$ *(Calibration)*

Theorem 1.1 (Cost Uniqueness [1]). *The unique continuous solution is $J(x) = \frac{1}{2}(x + x^{-1}) - 1$, satisfying $J(x) \geq 0$ (equality iff $x = 1$) and $J(x) = J(x^{-1})$.*

Proof sketch. Setting $H(t) = 1 + J(e^t)$ transforms (A2) into the d'Alembert equation $H(s+t) + H(s-t) = 2H(s)H(t)$ with $H(0) = 1$, $H''(0) = 1$. By the Aczél classification, $H(t) = \cosh t$, giving $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. □

The golden ratio $\varphi = (1 + \sqrt{5})/2$ emerges as the unique positive root of $x^2 = x + 1$ from the self-similarity of the discrete ledger. The fundamental RS constants are the voxel size ℓ_0 and tick time τ_0, with $c = \ell_0/\tau_0$ and $\hbar = E_{\text{coh}}\tau_0$ where $E_{\text{coh}} = \varphi^{-5}$ (RS-native).

The RS approach has four main results:

- GR = large-scale limit of the ledger dynamics
- No separate quantization of gravity is needed (no graviton)
- The EFE coupling κ is derived from the recognition wavelength
- Galaxy rotation curves follow from ILG without dark matter

2 Derivation of the Einstein Field Equations

2.1 RS Action

The RS action for geometry is:

$$S_{\text{RS}}[g] = \int \frac{1}{2\kappa}\sqrt{|g|}(R - 2\Lambda)\, d^4x + S_{\text{matter}} \tag{1}$$

This is the Hilbert action, but in RS the coupling κ is derived:

$$\kappa = \frac{8\pi G}{c^4} = \frac{\pi\lambda_{\text{rec}}^2}{\hbar c} \tag{2}$$

where λ_{rec} is the RS recognition wavelength — the curvature extremum of J.

2.2 EFE from Stationarity

Proposition 2.1 (EFE Emergence — Structural Identification)**.** *Varying S_{RS} with respect to $g^{\mu\nu}$ yields the Einstein Field Equations:*

$$\frac{\delta S_{\text{RS}}}{\delta g^{\mu\nu}} = 0 \implies G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu} \tag{3}$$

Structural argument. The standard variational derivation of the EFE from the Hilbert action uses:

- $\delta\sqrt{|g|}/\delta g^{\mu\nu} = -\frac{1}{2}\sqrt{|g|}g_{\mu\nu}$ (Jacobi formula)
- $\delta R/\delta g^{\mu\nu} = R_{\mu\nu}$ (Palatini identity, assuming torsion-free connection)
- Matter stationarity: $\delta S_{\text{matter}}/\delta g^{\mu\nu} = -\frac{1}{2}\sqrt{|g|}T_{\mu\nu}$

Combining these gives the EFE (see e.g. standard GR references [4]). The RS-specific content is not the variational calculation (which is identical to GR) but the claim that the RS action S_{RS} takes the Hilbert form with κ derived from RS constants. The derivation that the RS ledger dynamics produces the Hilbert action—rather than a more general action—is an identified open problem. □

3 Newton's Constant from RS

3.1 The Recognition Wavelength

The RS recognition wavelength λ_{rec} is the curvature extremum of J:

$$\frac{d^2}{dx^2}J(x)\bigg|_{x=\lambda_{\text{rec}}/\ell_0} = 0 \tag{4}$$

For $J(x) = \frac{1}{2}(x + x^{-1}) - 1$: $J''(x) = x^{-3}$, which is monotonically decreasing — the extremum is at $x \to \infty$ (the large-scale recognition horizon).

The physical interpretation: $\lambda_{\text{rec}} \sim c\tau_0 = \ell_0$ (the voxel scale).

3.2 Newton's Constant

The standard RS derivation (see companion paper on Einstein Field Equations) uses the quantum-gravitational duality $G \cdot \hbar = 1$ to obtain $G = 1/\hbar = \varphi^5$ in RS-native units. The recognition-wavelength approach gives:

$$G = \frac{c^4\kappa}{8\pi} = \frac{c^3\lambda_{\text{rec}}^2}{8\hbar} \tag{5}$$

With $\lambda_{\text{rec}} = \ell_0$ and $\hbar = E_{\text{coh}}\tau_0 = \varphi^{-5}$ (RS-native units $\ell_0 = \tau_0 = 1$):

$$G = \frac{\ell_0^2}{8\hbar} = \frac{1}{8\varphi^{-5}} = \frac{\varphi^5}{8}. \tag{6}$$

Remark 3.1 (Inconsistency with EFE Derivation). The recognition-wavelength derivation gives $G = \varphi^5/8$ while the quantum-gravitational duality $G \cdot \hbar = 1$ gives $G = \varphi^5$. These differ by a factor of 8. The resolution is that $\lambda_{\text{rec}} = \ell_0$ is not the correct identification: to recover $G = \varphi^5$, one needs $\lambda_{\text{rec}} = 2\sqrt{2}\,\ell_0$ (since $G = c^3\lambda^2/(8\hbar) = 8\ell_0^2/(8\varphi^{-5}) = \varphi^5$ when $\lambda_{\text{rec}} = 2\sqrt{2}\,\ell_0$). This gives $\ell_{\text{Pl}} = \sqrt{G\hbar/c^3} = \sqrt{\varphi^5 \cdot \varphi^{-5}} = 1 = \ell_0$, consistent with the EFE paper's identification $\ell_0 = \ell_P$. The recognition-wavelength approach is an alternative motivation for $G = \varphi^5$ that requires the identification $\lambda_{\text{rec}} = 2\sqrt{2}\,\ell_0$; the structural derivation of this identification is an open problem. The preferred RS value is $G = \varphi^5$ from $G \cdot \hbar = 1$.

4 Equivalence Principle

4.1 Why $m_i = m_g$ in RS

In RS, all masses arise from J-cost:

$$m = \frac{E_{\text{RS}}}{c^2} = \frac{J(x_{\text{config}}) \cdot E_{\text{coh}}}{c^2} \tag{7}$$

Both inertial mass (resistance to acceleration: $F = ma$) and gravitational mass (coupling to curvature: $T^{\mu\nu}$) derive from the same J-cost:

- Inertial: $m_i = \delta S/\delta a^\mu$ (variation with respect to acceleration)
- Gravitational: $m_g = \delta S/\delta g_{\mu\nu}$ (variation with respect to metric)

In RS, both variations give the same J-cost expression.

Proposition 4.1 (Equivalence Principle — Structural Identification)**.** *For any mass arising from J-cost in the RS framework:*

$$\frac{m_i}{m_g} = 1 \quad \textit{(identically).} \tag{8}$$

Structural argument. In RS, both inertial mass (m_i, resistance to acceleration) and gravitational mass (m_g, coupling to curvature) have the same origin: the J-cost of the particle's configuration on the φ-ladder, $m = J(x_{\text{config}}) \cdot E_{\text{coh}}/c^2$. Since there is only one mass—not two independent quantities—their ratio is identically unity. The claim that the RS variation $\delta S/\delta a^\mu$ and $\delta S/\delta g_{\mu\nu}$ are both equal to the same J-cost mass is the structural identification; a rigorous derivation showing both variations produce the same formula is an identified open problem. □

5 No Graviton

5.1 Gravity as Emergent Curvature

In RS, gravity is not mediated by an exchange particle (graviton). Instead:

- Gravity = curvature of the ledger metric
- The "gravitational force" is a geodesic deviation in the curved ledger
- There is no separate quantum of gravity to exchange

Remark 5.1 (No Graviton — Structural Observation). In RS, any gravitational interaction does not require a mediating fundamental particle. The gravitational coupling κ is a property of the emergent ledger metric, not of a boson exchange. Gravitons (if observed) would be collective excitations of the ledger metric, analogous to phonons in a crystal lattice—emergent but not fundamental. Detection of an elementary spin-2 graviton (irreducible to collective ledger dynamics) would challenge the emergent-gravity interpretation.

This is analogous to why there is no "phonon" in solid-state physics — phonons are collective excitations of the lattice, not fundamental particles. Similarly, "gravitons" (if they exist) are collective excitations of the ledger metric.

6 Galaxy Rotation Curves: ILG

6.1 The ILG Formula

The Inherent Lattice Gravity (ILG) formula for galaxy rotation:

$$v_c^2(r) = \frac{GM(<r)}{r}\left(1 + \alpha_t \cdot \frac{r}{r_0}\right) \tag{9}$$

where $\alpha_t = 0.5(1 - \varphi^{-1}) \approx 0.191$ is the ledger coupling and r_0 is a characteristic scale associated with the galaxy's baryonic scale radius (the half-mass radius of the baryonic distribution). In MOND-like language, $r_0 = \sqrt{GM(<r)/a_0}$, where $a_0 \approx 1.2 \times 10^{-10}$ m/s^2 is the RS-derived MOND acceleration scale.

The term $\alpha_t \cdot r/r_0$ grows with r, giving flat rotation curves at large radius without dark matter halos. This is the RS derivation of MOND-like phenomenology.

6.2 The α_t Value

Proposition 6.1 (α_t from φ)**.** *The ILG coupling parameter is the zero-parameter prediction:*

$$\alpha_t = \tfrac{1}{2}(1-\varphi^{-1}) = \tfrac{1}{2}\cdot\frac{\varphi-1}{\varphi} = \frac{1}{2\varphi^2} \approx 0.191.$$

Proof. From $\varphi^2 = \varphi + 1$ we get $\varphi^{-1} = \varphi - 1$, so $1 - \varphi^{-1} = 2 - \varphi$. To verify $2 - \varphi = \varphi^{-2}$: $(2-\varphi)(\varphi^2) = 2\varphi^2 - \varphi^3 = 2(\varphi+1) - \varphi(\varphi+1) = 2\varphi + 2 - \varphi^2 - \varphi = \varphi + 2 - (\varphi+1) = 1$. Hence $2 - \varphi = \varphi^{-2}$ and $\alpha_t = (1/2)(2-\varphi) = 1/(2\varphi^2) \approx 1/5.236 \approx 0.191$. □

Remark 6.2 (MOND connection and a_0). The MOND acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m/s^2 in RS is not a free parameter but emerges from the characteristic scale r_0 of the baryonic distribution: $a_0 = GM(< r_0)/r_0^2$, which defines r_0 as the radius where the Newtonian acceleration equals a_0. The RS prediction is that all galaxies should fall on a single universal ILG relation with the same $\alpha_t = 1/(2\varphi^2)$ (no per-galaxy fitting). This is consistent with the SPARC galaxy survey data [3]. A rigorous derivation of the ILG kernel from the RS ledger vertices—and the consequent prediction of a_0 from RS native constants—is an identified open problem.

7 Gravitational Waves at c

7.1 Why GW Speed = c Exactly

Gravitational waves propagate as ripples in the ledger metric. The speed of propagation in the ledger is:

$$c_{\mathrm{GW}} = \frac{\ell_0}{\tau_0} = c \tag{10}$$

because both are defined by the same fundamental ledger quantities ℓ_0 (voxel size) and τ_0 (tick time).

Proposition 7.1 (Gravitational Wave Speed)**.** *GW speed = c exactly in RS — not a coincidence but a structural identity.*

Confirmed by GW170817: $|c_{\mathrm{GW}}/c - 1| < 10^{-15}$.

8 Predictions

1. κ_{RS}: Newton's constant $G = c^3\ell_0^2/(8E_{\mathrm{coh}})$.
2. **No graviton**: No spin-2 boson detected at any energy up to M_{Pl}.
3. **Equivalence principle**: $m_i/m_g = 1$ to all orders; no deviation expected.
4. **ILG rotation curves**: $v_c^2 \propto r$ at large r for isolated galaxies, with universal $\alpha_t \approx 0.191$.
5. **GW speed**: $c_{\mathrm{GW}} = c$ exactly, to all orders.

9 Conclusion

RS derives all of gravitational physics from the ledger:

1. EFEs from action stationarity with coupling $\kappa = \pi\lambda_{\text{rec}}^2/(\hbar c)$
2. Equivalence principle as a theorem (all masses from J-cost)
3. No graviton — gravity is emergent ledger curvature
4. Flat rotation curves from ILG with $\alpha_t = 0.5(1 - \varphi^{-1})$
5. GW speed $= c$ exactly (same ledger defines both)

References

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